

Thomas VAUDESCAL

Quantitative Developer | Quantitative Researcher

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Profile

Quantitative developer and researcher building research-to-production trading systems in **Python, Numba and Rust**: an event-driven options backtester, an exotics pricing engine with Greeks via AAD, and a **live systematic SPX options strategy**. MSc in Financial Engineering, HEC Montréal (GPA **3.88/4.3**, thesis graded **A+**).

Education

HEC Montréal | MSc in Financial Engineering | GPA 3.88/4.3

Montréal | August 2022 - December 2024

- **Thesis, A+ with commendation from the examination committee:** Gamma Exposure and Intraday Volatility Dynamics in the SPX Options Market. Research on dealer hedging and intraday volatility dynamics, subsequently applied to a live systematic options strategy.
- **Scholarships of excellence:** Deloitte M.Sc., Alpha Fixe Capital, Jocelyne & Jean C. Monty I, and Jocelyne & Jean C. Monty II.
- **Competitions:** National Bank challenge winning team at ConUHacks VIII (2024), top 10 of 66 teams at the McGill FIAM Portfolio Management Hackathon, and 2nd place at the HEC Montréal Trading Club Trading Competition.
- **Coursework:** Stochastic Calculus, Derivatives Pricing, Financial Econometrics, Numerical Methods, Machine Learning for Finance.

HEC Montréal | BBA in Mathematics, Finance & Economics

Montréal | August 2017 - May 2022

- **GPA:** 3.8/4.3 for specialization courses. Quantitative track combining mathematics, economics and finance.

Université de Montréal | Minor in Mathematics

Montréal | August 2018 - August 2022

- Alongside the BBA: probability, mathematical statistics, stochastic processes, linear algebra, real analysis, C/C++.

Professional Experience

Synerqo | Founder & Quantitative Researcher

Montréal | May 2025 - Present

- Built an in-house **event-driven options backtesting engine in Python and Numba** (spreads, slippage, commissions, margin, assignment), validated against QuantConnect. Runs one month of SPX intraday option chains (3.7M rows) in about 40 ms, 4x to 7x faster than the same code with Numba disabled, with identical results.
- Wrote **from scratch an exotic options and structured products pricing engine** with Monte Carlo and PDE solvers, Greeks via **Adjoint Algorithmic Differentiation (AAD)** on JAX, and Heston, SABR and SVI calibration. Greeks cost about 3x one price (GBM, 500k paths) and run 2.9x faster than bump-and-reprice, within 0.25% of Black-Scholes.
- Developed **Rust execution components** for market-data ingestion, quote updates and order routing, and profiled their latency against the Python research stack.
- Designed, backtested and **deployed a systematic SPX options strategy to live trading**, turning MSc research on dealer gamma exposure into a production signal and execution workflow.
- Prototyped systematic options market making with skew-aware quoting, inventory-based price adjustments and microstructure-based adverse-selection filters.

Deloitte | Quantitative Analyst, Financial Risk

Montréal | October 2024 - June 2025

- Developed and validated **credit risk models (PD, LGD, EAD)** for major Canadian financial institutions under IFRS 9 and Basel III/IV frameworks.
- Led independent model validation engagements, assessing assumptions, calibration and discriminatory power through statistical backtesting, sensitivity analysis and stress testing.
- Built **automated monitoring dashboards and validation pipelines in Python**, reducing manual review time by **40%**.
- Authored validation reports and regulatory documentation, presented to **senior stakeholders and client risk committees** (OSFI context).

National Bank Financial Markets | Quantitative Developer Intern

Montréal | May 2023 - December 2023

- In Global Equity Derivatives R&D, developed and deployed low-volatility and momentum **SmartBeta strategies covering \$800M in AUM** within a \$7B structured products division, with constrained portfolio optimization (cvxpy).
- Built an **end-to-end Python backtesting framework**: signal generation, portfolio construction, transaction costs, risk analytics and performance attribution.
- Delivered **\$1M in annual cost savings** through strategy optimization and migration off an external vendor.
- Worked with **structuring and trading teams** to integrate quantitative signals into derivatives pricing and hedging workflows, and delivered performance and risk reports to the trading desk and portfolio managers.

- Led **consulting engagements in insurance analytics**: data cleaning, exploratory analysis, predictive modeling and automated reporting to support client decision-making.
- Designed and delivered a **Python, statistics and machine learning curriculum** for Videns Academy, adopted by UQAC for a university course. Promoted from intern to full-time Data Analyst.

Projects

Quant Paper Lab | github.com/Thomas4390/quant-paper-lab

Open source | 2026

- Landmark finance papers reproduced on public data with interactive Streamlit figures (Jegadeesh and Titman 1993: published 1.31% monthly momentum return). A test cuts network access to enforce offline computation. Diagnosed a native pyarrow segfault.

FX Strategies | github.com/Thomas4390/fx_strategies

Open source | 2026

- Research framework for intraday and daily FX strategies on VectorBT Pro: parameter grids, purged walk-forward cross-validation and Numba metric kernels. 46 tests, including numerical-equivalence checks (rtol 1e-10) against frozen baselines.

Ganesha | ganesha.synerqo.com

Synerqo | Production

- Production RAG assistant for professional services firms (Python, FastAPI, Qdrant, Next.js): retrieval, reranking, answers linked to their sources, and an LLM evaluation framework with claim-level evidence checks.

Academic Involvement

Trading Club | Vice President Research

August 2023 - May 2024

- Managed a team of 4 students reproducing research papers on coastline trading (high-frequency FX strategy) and energy trading (modeling DA-RT electricity price spikes). Developed and backtested the strategies in Python.

BNI-HEC Investment Fund | Quantitative Analyst

September 2022 - May 2023

- R&D team of a \$5M student-managed portfolio. Developed quantitative investment models for strategic and tactical portfolios, and built the Fund's dashboards, reports, performance attribution and risk analysis.

Data Science Committee | Vice President Education

August 2020 - August 2021

- Organized the "AI in 2020" conference (40+ students and industry professionals), a Lunch and Learn on sentometrics research, Python workshops and machine learning seminars.

Teaching Assistant | Intermediate Microeconomics, Mathematics Preparatory Year

August 2021 - May 2023

- Corrected exams and tutorials, held consultation hours and created exercises for a course of 200+ students.

Technical Skills & Interests

Programming	Python, C++, Rust, SQL, R, MATLAB.
Computer Science	Data structures and algorithms, object-oriented design, event-driven systems, concurrency (asyncio), latency profiling, performance optimization.
Python Stack	NumPy, Numba, JAX, pandas, SciPy, statsmodels, scikit-learn, PyTorch, cvxpy, QuantLib.
Quant Finance	Options and structured products pricing, Greeks via AAD, Monte Carlo simulation, PDE methods, volatility calibration (Heston, SABR, SVI), backtesting, portfolio optimization, market microstructure.
Risk Modeling	Credit risk (PD, LGD, EAD), model validation, statistical backtesting, sensitivity analysis, stress testing, model monitoring.
Data & Tools	Bloomberg Terminal & API, AWS (EC2, S3), Docker, Git, Linux/Shell, REST APIs, WebSocket, large-scale data processing.
Web & AI	TypeScript, React, Next.js, FastAPI, PostgreSQL, RAG, LLM integration, Qdrant.
Languages	French (native), English (professional proficiency).
Interests	Chess (2000 Elo, Chess.com rapid), mathematics, philosophy, programming. Marathon running, track and field, mountain hiking, indoor climbing, swimming, scuba diving.